



C20-EC-504

7643

BOARD DIPLOMA EXAMINATION, (C-20)

MARCH/APRIL—2025

DECE – FIFTH SEMESTER EXAMINATION

INDUSTRIAL ELECTRONICS

Time : 3 Hours]

[Total Marks : 80

PART—A

3×10=30

Instructions : (1) Answer **all** questions.

(2) Each question carries **three** marks.

(3) Answers should be brief and straight to the point and shall not exceed five simple sentences.

1. List any three applications of DIAC.
2. Draw the volt-ampere characteristics of UJT.
3. Draw the symbols of SCS, SBS and SUS.
4. List the applications of opto coupler.
5. Distinguish between LED and photodiode.
6. What is magnetostriction effect?
7. Write the working principle of accelerometer.
8. Classify different industrial heating methods.
9. List different types of electrical welding.
10. Write the working principle of inverter.

PART—B

8×5=40

- Instructions :** (1) Answer **all** questions.
(2) Each question carries **eight** marks.
(3) Answers should be comprehensive and criterion for valuation is the content but not the length of the answer.

11. Explain the construction and working of SCR with characteristics.

(OR)

Explain the construction and working of TRIAC with characteristics.

12. Explain the construction and working of photo diode and draw its characteristics.

(OR)

Explain the construction and working of photovoltaic cell and write its applications.

13. Explain the construction and working of Thermocouple transducer.

(OR)

Explain the construction and working of pulsed echo ultrasonic flaw detector.

14. Explain about HF power source for induction heating.

(OR)

Explain the principle of dielectric heating and write its applications.

15. Explain the working of SMPS with block diagram.

(OR)

Explain the working of offline UPS and online UPS.

PART—C

10×1=10

- Instructions :**
- (1) Answer the following question.
 - (2) The question carries **ten** marks.
 - (3) Answer should be comprehensive and the criterion for valuation is the content but not the length of the answer.

16. Explain open loop and closed loop control systems with examples.

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